KEREM DELIKOYUN

AI Engineer

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SUMMARY

AI Engineer at GIC with over 10 years of experience architecting and operating production-grade AI platforms and agentic systems on cloud-native infrastructure. Doctor of Engineering (Dr.-Ing.) from TU Munich with deep expertise in agent runtime design, multi-agent orchestration, distributed systems, and full-stack AI platform engineering.

Currently building GIC's foundational AI agent platform and agent harness, the runtime that lets teams across the organisation build and run secure, production-grade AI agents for finance and asset management at scale. Previously led an international team at TUMCREATE to deliver a patented health AI platform, secured commercialization funding, and developed multiple open-source agent harnesses for scientific discovery.

EXPERIENCE

GIC | AI Engineer, AVP

Sep 2026 - Present | Singapore

- Building GIC's foundational enterprise AI agent platform, enabling teams across the organisation to develop and operate secure, production-grade AI agents for investment, research, and operations workflows in asset management.
- Designing the agent harness and runtime: multi-step workflow orchestration, tool and memory management, durable execution, and scaling of agent workloads, built to the security and reliability standards of a financial institution.

TUMCREATE | Research Fellow, Team Lead

Jan 2026 - Aug 2026 | Singapore

- Led a multidisciplinary team of engineers and researchers across Singapore and Germany, owning the end-to-end product lifecycle: architecture and technical roadmaps, specifications and API contracts, standards for code review, system design, and testing, and cross-functional delivery using agile practices.
- Architected a production-grade Agentic OS for automated knowledge discovery and data analysis as a full-stack application (FastAPI, React) with RAG on PostgreSQL and Vector DBs, short/long-term memory, skills/plugins/MCP integration, and unit, integration, and E2E tests with CI/CD (GitHub Actions).
- Improved domain-specific reasoning by multi-GPU distributed post-training of LLMs (Qwen 3.5) on proprietary datasets and integrated the fine-tuned models into the agent runtime with model routing, structured output parsing, and provider failover, instrumented end-to-end with monitoring and evaluation dashboards (MLflow, Opik, Grafana).

TUMCREATE | Research Associate

Aug 2021 - Dec 2025 | Singapore

- Built an end-to-end LLM-based multi-agent system (TriAgent) for automated data analysis and deep research with RAGAS and Opik LLM-as-judge evaluation and reliability patterns (retries with jitter, timeouts, idempotent steps).
- Reduced manual analysis effort by >95% (~3 days to 15 min) at ~500 s and <$1 per run on AWS, with end-to-end logging, distributed tracing, and observability across the runtime (GitHub Repo | Paper).
- Engineered a production AI pipeline (CNN, GAT, ViT) on multi-GPU infrastructure with PyTorch, achieving >70x throughput improvement (2.5 h to 2 min) and >99% storage reduction per sample (30 GB to 15 MB) across 100,000+ data points (>15 TB), with real-time inference at <10 ms/frame and ~$0.01/inference, distributed tracing, and SLA monitoring (GitHub Repo | Paper).
- Secured commercialization funding from SMART and NRF for the patented AI deep-tech solution by articulating technical vision and trade-offs to investment stakeholders (News).

Izmir Institute of Technology | Research Assistant

Sep 2016 - Jul 2021 | Türkiye

- Enabled automated analysis of imaging data by building a CNN-based recognition pipeline with end-to-end data processing workflows from raw sensor data through data curation, feature engineering, model training, selection, and deployment (Paper).
- Designed and implemented feature extraction and model evaluation workflows for troubleshooting and reliability testing, earning 4th place in the National Patent Competition for the algorithms and software architecture.

KEY SYSTEMS BUILT

- **Agentic OS:** Production-grade full-stack AI platform (FastAPI/React/PostgreSQL/AWS) developed with AI-native CLI coding agents (Claude Code, Cursor, OpenClaw), with RAG, persistent memory, fine-tuned LLMs, MCP integration, evaluation dashboards (MLflow, Opik, Grafana), and CI/CD for repeatable deployments.
- **TriAgent Multi-Agent System:** LLM-based multi-agent system for automated data analysis and deep research with RAGAS/Opik evaluation (topic adherence 55.7%, faithfulness 0.42), reliability patterns (retries with jitter, timeouts, idempotent steps), and observability. >95% effort reduction at <$1/run on AWS (GitHub Repo).
- **Organon:** Open-source Agentic OS for automated scientific discovery, with a multi-agent runtime (researcher → writer → verifier → reviewer with isolated context), 3 MCP server integrations, 31 modular skills, durable cross-session memory, cron + watchdog scheduling, and PreToolUse verification hooks for citation/claim integrity (Website | GitHub | Substack).

SKILLSET

Technical: Python, FastAPI, gRPC, JavaScript/React | PyTorch, HuggingFace, Unsloth, vLLM, Ray | LangChain, LangGraph, AutoGen, OpenAI Agents SDK, MCP, RAGAS, Opik | Workflow Orchestration & Durable Execution (Prefect, Temporal patterns, cron + watchdog), Event-Driven Architecture (Kafka, SQS/SNS, Redis Streams) | PostgreSQL, Vector DB (Pinecone, Chroma, FAISS), dbt, Spark | MLflow, Langfuse, OpenTelemetry, Grafana | AWS (EKS, Bedrock, SageMaker, Lambda, DynamoDB), Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, ArgoCD) | AI-native Dev Tools (Claude Code, Cursor, OpenClaw)

Domain: Agent Runtime Architecture, Multi-Agent Systems & Failure Modes, Agentic Memory & Context Engineering, MCP Servers/Registries, RAG, SFT/DPO/RLHF/PEFT, Multi-GPU/Distributed Post-Training, LLM Evaluation Frameworks, Distributed Systems, Resilience Patterns (circuit breakers, retries with jitter, dead-letter queues, idempotency), MLOps/LLMOps/AgentOps, Experiment Tracking & Observability

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EDUCATION

Technical University of Munich (TUM)
Doctor of Engineering (Dr.-Ing.) - Electrical and Computer Engineering
2023 - 2025 | *Germany*

Technical University of Munich (TUM)
Master of Science (M.Sc.) - Physics, graduated with the highest distinction
2022 - 2023 | *Germany*

Gaziantep University
Bachelor of Science (B.Sc.) - Engineering Physics, graduated as salutatorian
2009 - 2015 | *Türkiye*

CERTIFICATIONS & AWARDS

- SMART IG 2.0 Commercialization Grant by SMART & NRF (Sep 2025 | Singapore)
- 1st Place, TextArena Agent Hackathon by AITinkerers (Mar 2025 | Singapore)
- 4th Place in National Patent Competition by Turkish Patent Institute (Oct 2020 | Türkiye)
- Amazon Web Services: AWS Certified AI Practitioner & AWS Certified Cloud Practitioner (Oct 2024 | Singapore)
- DataTalksClub: LLMOps | MLOps Certification (Sep/Oct 2024 | Singapore)

SELECTED PATENTS & PUBLICATIONS

- Quantitative Analysis of Immune Cell and Platelet Aggregate Activation Using Real-Time and End-To-End Image Processing Framework for Digital Holographic Microscopy - 10202500263R (pending), 2025, Singapore
- Real-Time And End-To-End Image Processing Framework for Digital Holographic Microscopy - 10202401892U (Patent), 2024, Singapore
- K. Delikoyun et al., "TriAgent: Automated Biomarker Discovery with Deep Research Grounding for Triage in Acute Care by LLM-Based Multi-Agent Collaboration," arXiv, 2025 (Paper)
- K. Delikoyun et al., "RT-HAM: Real-Time Holographic Agent Microscopy," CVPR 2025 Workshop, 2025 (CVPR 2025)